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MODEL SELECTION AND VALIDATION REPORT ( 2017_2022_biomass model )
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## OBSERVATIONS:954

## SANITY: OK − all models in the selection process passed sanity checks.

## SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

## MODEL SUMMARY ( 2017_2022_biomass model )

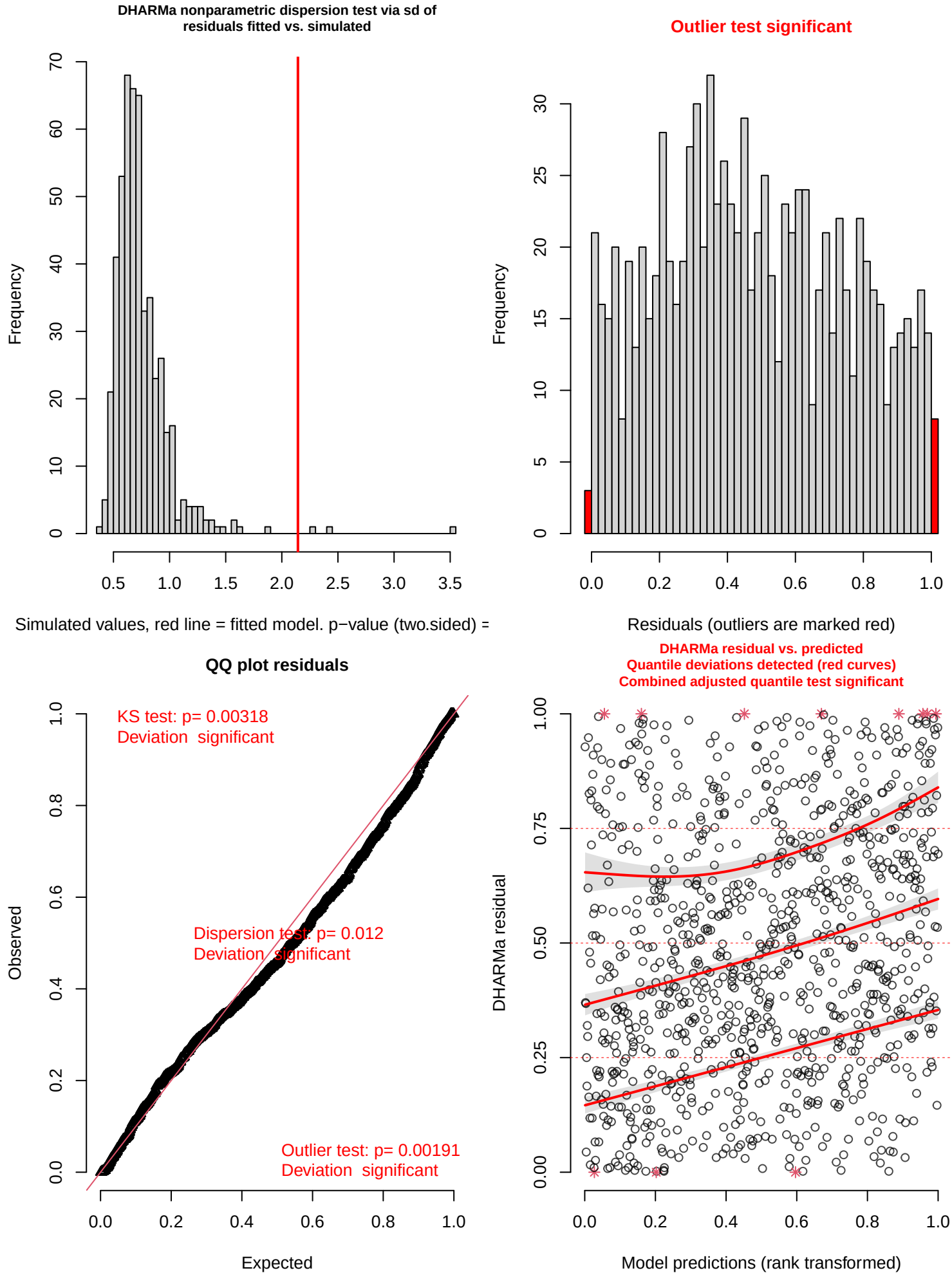
Spatiotemporal model fit by REML ['sdmTMB']
Formula: fixed_formula
Time column: year
Family: lognormal(link = 'log')

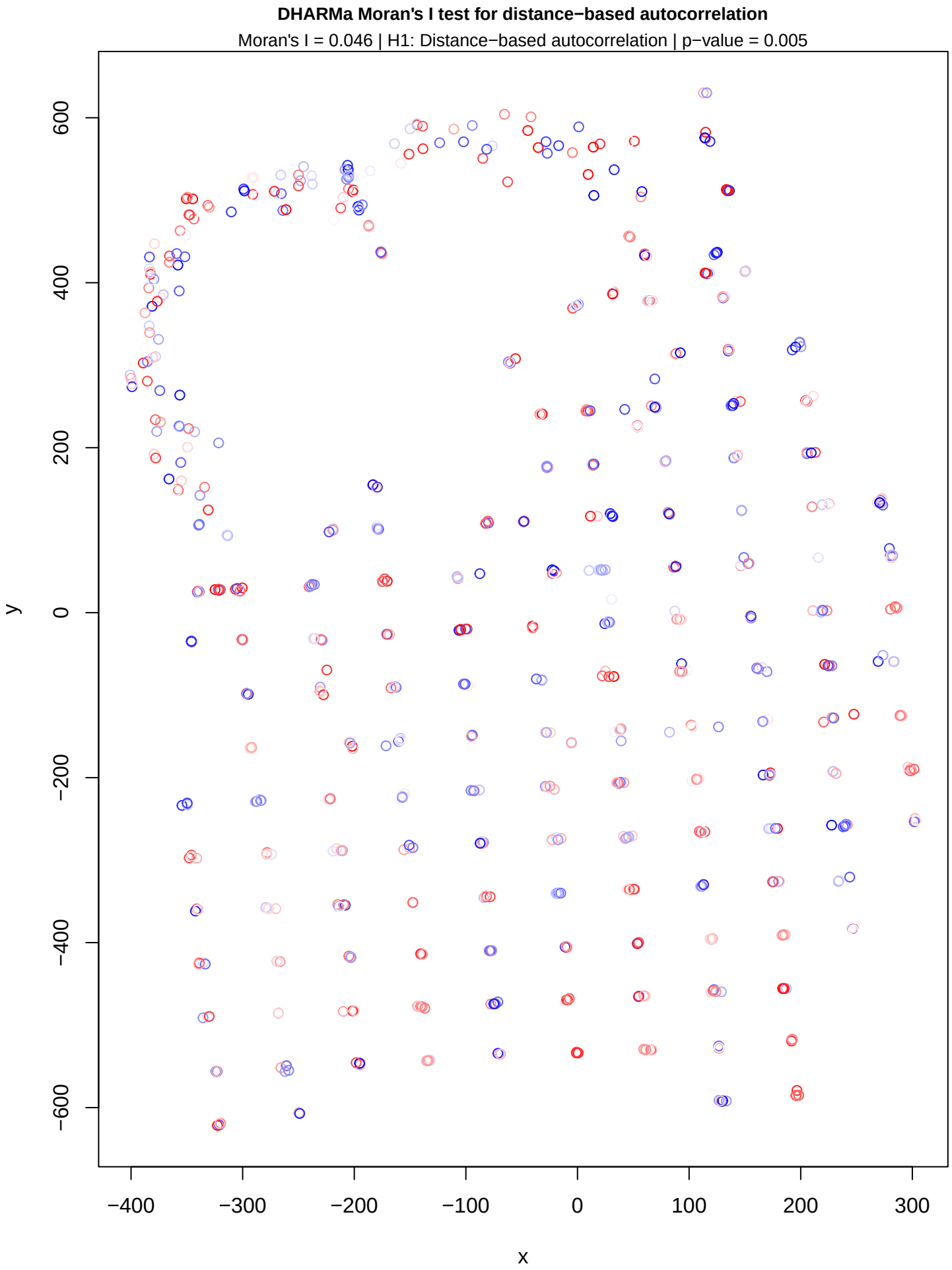
Conditional model:
              coef.est coef.se
(Intercept)       7.22    0.15
kde_fric          0.05    0.03
kde_feve         −0.10    0.03
tric             −0.02    0.03
teve             −0.28    0.03
sbt_mean          0.33    0.09
sbt_sd           −0.03    0.05
sst_sd            0.01    0.05
sic_mean         −0.11    0.08
log_chla_mean     0.10    0.05
depth            −0.28    0.05

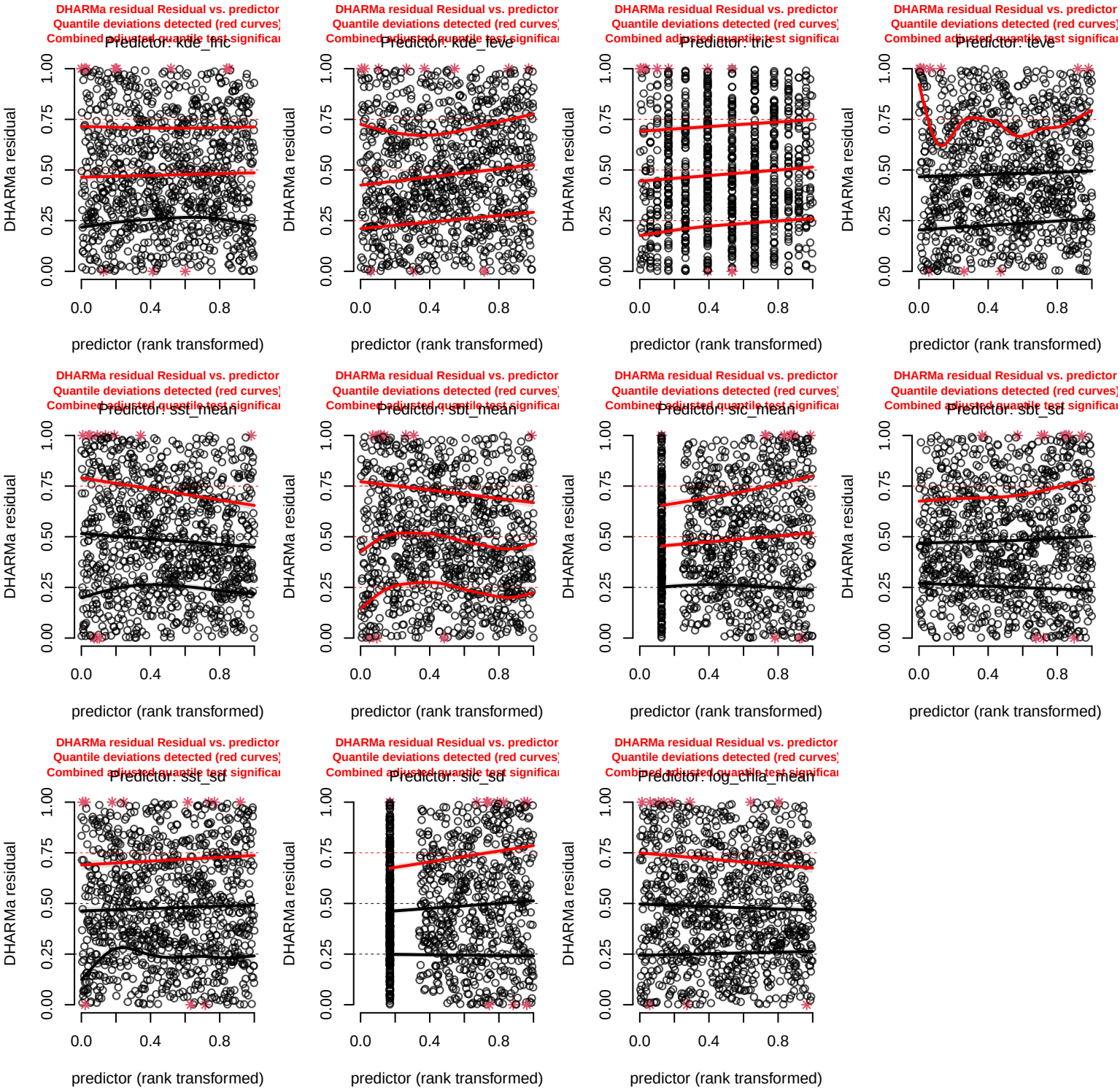
Dispersion parameter: 0.74
Spatiotemporal AR1 correlation (rho): 0.74
Matérn range: 291.46
Spatiotemporal marginal AR1 SD: 0.56
REML criterion at convergence: 7838.923

See ?tidy.sdmTMB to extract these values as a data frame.

## SELECTION STATISTICS
                                                   df      AIC delta_AIC
spatiotemporal_random_fields_ar1                   15 15707.85  3.870755
spatiotemporal_random_fields_ar1_with_anisotropy 17 15711.72 29.198382
spatiotemporal_random_fields_iid                   14 15740.92  2.670305
spatiotemporal_random_fields_iid_with_anisotropy 16 15743.59 12.600766
spatial_random_fields                              14 15756.19  2.906982
spatial_random_fields_with_anisotropy              16 15759.09 61.817917
error_only                                         12 15820.91        NA
```


DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
Simulated values, red line = fitted model. p−value (two.sided) =
Outlier test significant
Frequency
Residuals (outliers are marked red)
QQ plot residuals
KS test: p= 0.00318
Deviation significant
Dispersion test: p= 0.012
Deviation significant
Outlier test: p= 0.00191
Deviation significant
Observed
Expected
DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
Model predictions (rank transformed)


DHARMa Moran's I test for distance−based autocorrelation
Moran's I = 0.046 | H1: Distance−based autocorrelation | p−value = 0.005
y
600
400
200
0
−200
−400
−600
x
−400
−300
−200
−100
0
100
200
300


DHARMa residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significa
Predictor: kde_fnc
Predictor: kde_feve
Predictor: tric
Predictor: teve
Predictor: sst_mean
Predictor: sbt_mean
Predictor: sic_mean
Predictor: sbt_sd
Predictor: sst_sd
Predictor: sic_sd
Predictor: log_chla_mean
DHARMa residual
predictor (rank transformed)
0.00
0.25
0.50
0.75
1.00
0.0
0.4
0.8